

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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CT
GROUP

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INDEX
NUMBER

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BIOLOGY

9744/04

Paper 4 Practical

13 Aug 2018

2 hours 30 minutes

Candidates answer on the Question Paper

Additional materials: As listed in the Confidential Instructions

READ THESE INSTRUCTIONS FIRST

Write your CT group, index number and name on all the work you hand in.

Give details of the practical shift and laboratory, where appropriate, in the boxes provided.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

Shift
Laboratory

For Examiner's Use	
1	24
2	17
3	14
Total	55

- 1 The enzyme amylase, **E**, hydrolyses (breaks down) starch, to a reducing sugar. You are required to investigate how much reducing sugar diffuses from a mixture of starch and amylase through a partially permeable wall of Visking tubing.

You are provided with:

labelled	contents	hazard	volume /cm ³
E	2.0% amylase solution	irritant	20
S	1.0% starch suspension	none	20
W	distilled water	none	150

labelled	contents	hazard	details	quantity
V	Visking tubing	none	15 cm length in distilled water	1

If **E** comes into contact with your skin, wash it off immediately under cold water.

It is recommended that you wear suitable eye protection.

Fig. 1.1 shows the apparatus you will set up for this investigation.

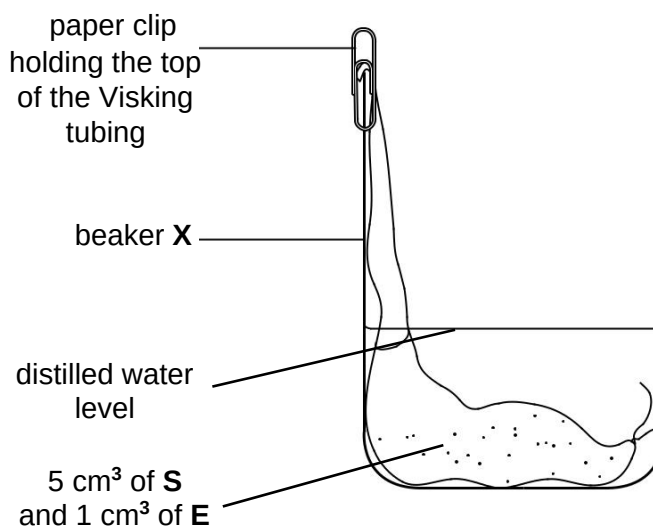


Fig. 1.1

Proceed as follows:

1. Tie a knot in the Visking tubing as close as possible to one end, so that it seals the end.
2. To open the other end, wet the Visking tubing and rub the tubing gently between your fingers.
3. Put 5 cm³ of **S** into the Visking tubing.
4. Put 1 cm³ of **E** into the Visking tubing.
5. Rinse the outside of the Visking tubing by dipping it into the water in the container labelled **V**.

Look carefully at Fig. 1.1. This has been set up so that the volume of water is as small as possible to cover the Visking tubing. The part of the Visking tubing containing the mixture is on the bottom of the beaker.

6. Put the Visking tubing into the beaker, labelled **X**, as shown in Fig. 1.1.
 7. Put **W** into the beaker up to the level shown on Fig. 1.1 using a syringe so that you can measure the volume of **W**.
- (a) (i) State the volume of **W** needed to reach the water level as shown in Fig. 1.1.

volume of **W** = cm³ [1]

8. Leave the apparatus for 15 minutes.

While you are waiting, continue with Question 1.

9. After 15 minutes, remove the Visking tubing and put it into the container labelled '**For waste**'.

You are required to:

- prepare a serial dilution of the 1.0% reducing sugar solution, **R1**
- carry out the Benedict's test for the known concentrations of reducing sugar and the water surrounding the Visking tubing
- use the results to estimate the concentration of reducing sugar in the water surrounding the Visking tubing.

You are provided with:

labelled	contents	hazard	volume / cm ³
W	distilled water	none	250
R1	1.0% reducing sugar solution	none	30
Benedict's	Benedict's solution	irritant	30

It is recommended that you wear suitable eye protection.

If **Benedict's** comes into contact with your skin, wash it off immediately with cold water.

[Turn over

- (ii) You are required to make a **serial** dilution of the 1.0% reducing sugar solution, **R1**, which reduces the concentration of reducing sugar solution by **half** between each successive dilution. You will also need to set up a control, **C**.

You are required to make up at least 10cm³ of each concentration of reducing sugar solution in the small glass vials provided.

Complete Table 1.2 to show how you will make the concentrations of the reducing sugar solutions, **R2, R3, R4 and R5**, and show how you will set up the control, **C**.

Table 1.2

	R1	R2	R3	R4	R5
Concentration of reducing sugar solution / %					
Label of reducing sugar solution to be diluted		R1			
Volume of reducing sugar solution to be diluted/ cm ³					
Volume of distilled water, W, to make the dilution/ cm ³					
<u>Description of the control, C:</u>					

10. Set up a boiling water-bath ready for step 12.
11. Prepare the concentrations of reducing sugar solution, as decided in **(a)(ii)**, in the glass vials provided.
12. Carry out the Benedict's test on each of the concentrations of reducing sugar solution and record your results in **(a)(iii)**.

You will need to use 2cm³ of each of the concentrations of reducing sugar solution with 3cm³ of Benedict's solution.

13. Test each solution **separately** and record in **(a)(iii)** the time taken for the **first** appearance of any colour change. If there is no colour change after 180 seconds record as 'more than 180'.

(iii) Prepare the space below and record your results.

[5]

(iv) Describe **one** significant source of error when carrying out steps 12 and 13.

.....
.....
..... [1]

Before proceeding, check that you have carried out step 9 on page 3.

[Turn over

14. Carry out the Benedict's test on a sample taken from beaker **X** (sample **X**) and record the time taken for the first colour change to appear.

- (v)** State the time taken for the first colour change for sample X.

time taken = [1]

- (vi)** Complete Fig. 1.3 to show:

- the positions of each of the percentage concentrations of reducing sugar solution
- an estimate of the concentration of reducing sugar in the sample **X**, using a letter **X**.

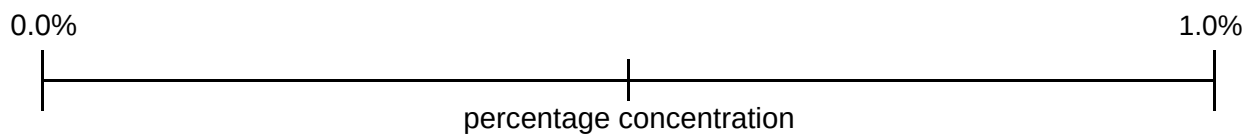


Fig. 1.3

- (vii) Describe how you could use this procedure to produce a more accurate estimate of the concentration of reducing sugar in the sample **X** than the one given in **(a)(vi)**.
Do **not** include the use of a colorimeter in your answer.

..... [3]

- (b)** A scientist investigated the effect of temperature on the activity of the enzyme in the Visking tubing.

All other variables were kept constant.

The quantity of reducing sugar diffusing through the wall of the Visking tubing was measured by a dye in the surrounding solution. The dye reacted with the reducing sugar. The more reducing sugar present the more intense the colour.

A colorimeter was used to measure the absorbance of light by the coloured solution. The absorbance of light by pure water is 0.00 arbitrary units.

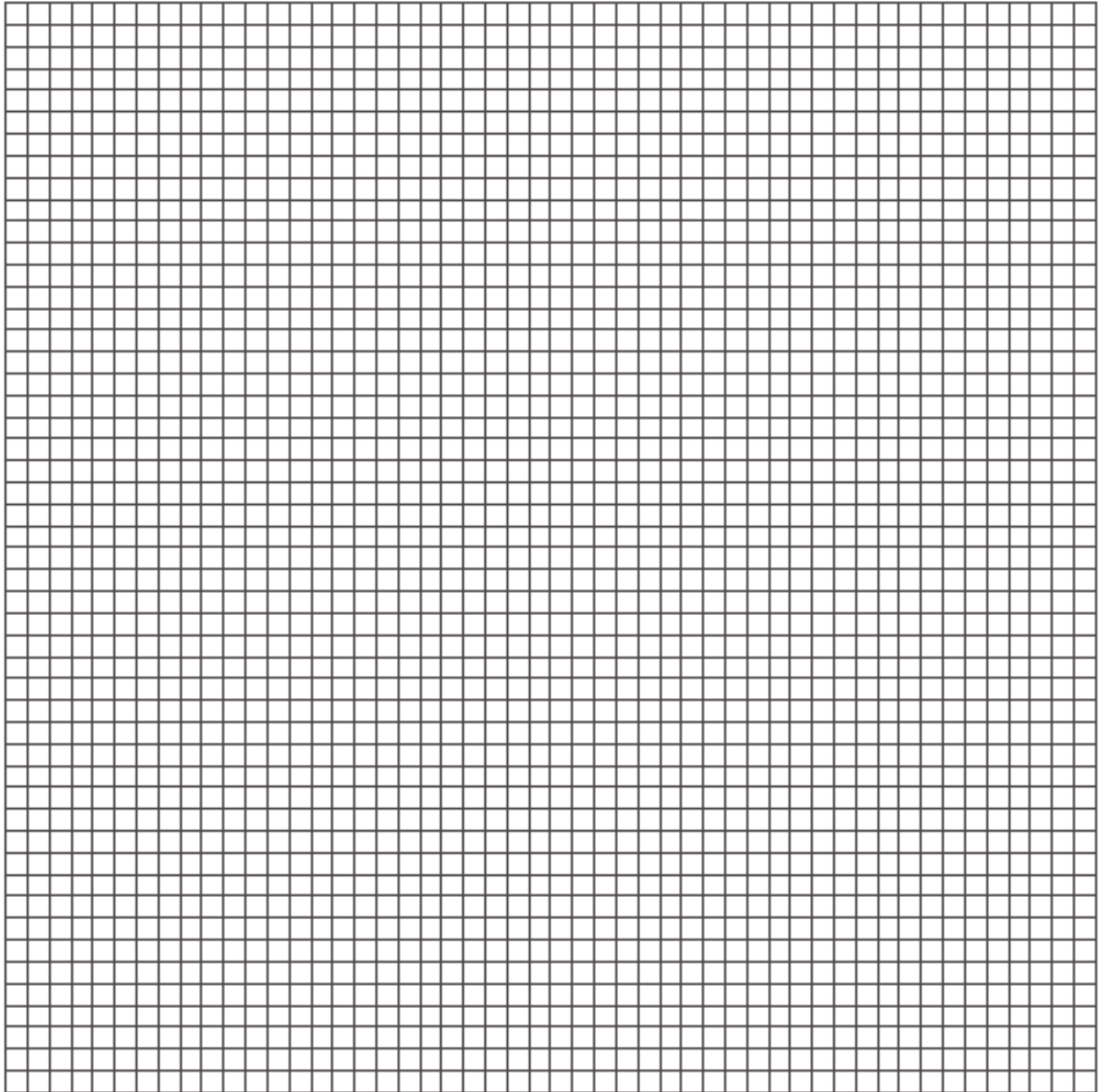
The results are shown in Table 1.4.

Table 1.4

temperature / °C	absorbance of light by the coloured solution / arbitrary units
30	0.90
41	1.46
49	1.58
59	1.10
70	0.65

Use a sharp pencil for graphs.

- (i) Plot a graph of the data shown in Table 1.4.



[4]

- (ii) Explain the difference in the absorbance of light between 49 °C and 70 °C.

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..... [3]

[Total: 24]

2 L1 is a slide of a stained transverse section through a plant leaf.

You are not expected to be familiar with this specimen.

You are required to:

- use the eyepiece graticule to measure part of a leaf and a vascular bundle
- use these measurements to calculate the depth of the vascular bundle as a percentage of the depth of the leaf
- draw a plan diagram of part of the leaf.

(a) The eyepiece graticule in the microscope can be used to measure different tissues. Select a part of the leaf on **L1** which shows the widest part of the leaf (mid-rib) shown by **Y** in Fig. 2.1.

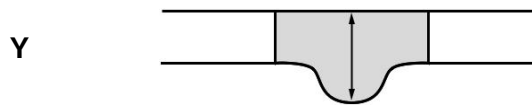


Fig. 2.1

(i) Use the eyepiece graticule in the microscope to measure:

- the depth of the leaf at **Y**
- the depth of the vascular bundle at **Y**.

depth of leaf eyepiece graticule units

depth of vascular bundle eyepiece graticule units [1]

(ii) Use the measurements from **(a)(i)** to calculate the depth of the vascular bundle as a percentage of the depth of the leaf.

You may lose marks if you do not show your working.

answer =% [2]

[Turn over

Use a sharp pencil for drawing.

- (iii) Use the measurements from **(a)(i)** to help you draw a large plan diagram of the section of the leaf shown by the shaded area in Fig. 2.2.

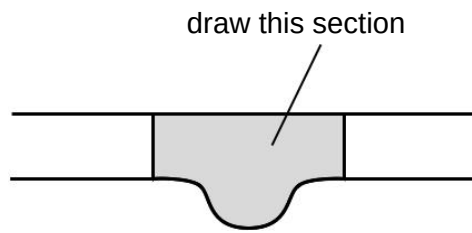


Fig. 2.2

You are expected to draw the correct shape and proportions of the different tissues.

Use **one** ruled label line and label to identify the vascular bundle.

- (iv) Observe the vascular bundle in the central part of the leaf on **L1**.
Select one group of **four** adjacent (touching) xylem vessel elements. Each element must touch at least one of the other elements.

Make a large drawing of this group of **four** xylem vessel elements.

Use **one** ruled label line and label to identify the lumen of **one** xylem vessel element.

- (b) Fig. 2.3 is a photomicrograph of a stained transverse section through a different type of leaf. You are not expected to be familiar with this specimen.

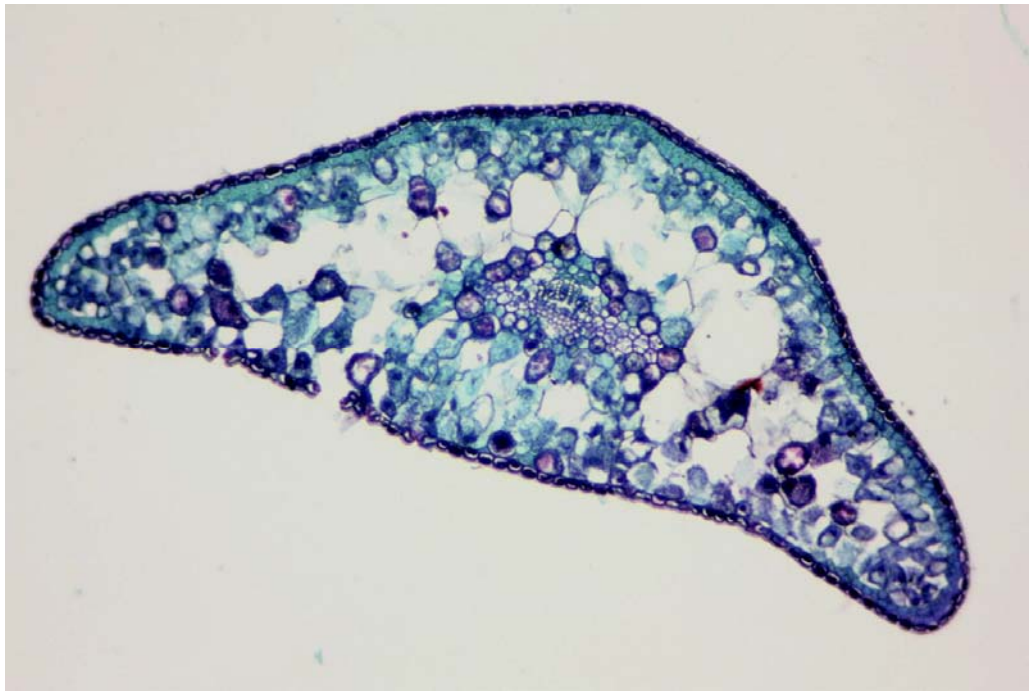
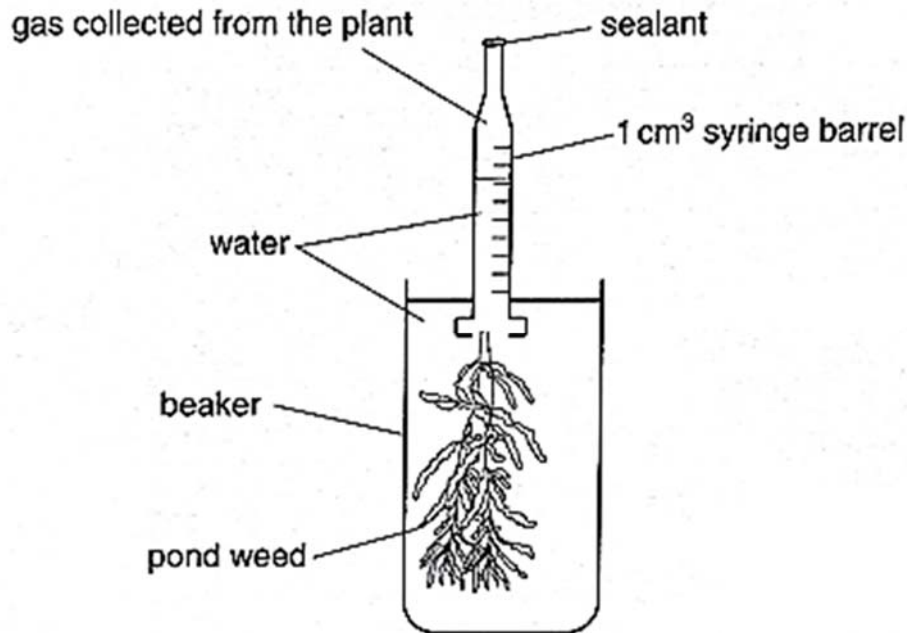


Fig. 2.3

Prepare the space below so that it is suitable for you to record the observable differences between the leaf sections on **L1** and in Fig. 2.3.

Record your observations in the space you have prepared.

- 3 The rate of photosynthesis can either be measured by the rate at which carbon dioxide is taken in or the amount of oxygen that is given out. Some water plants release bubbles of gas from a freshly cut stem when illuminated. Light intensity is controlled using five filters, F1, F2, F3, F4, F5.



Different water plants are adapted to different light intensities. A sun-loving water plant is adapted to high light intensities while a shade-loving water plant is adapted to low light intensities.

Using this information, the set-up above and your own knowledge, design an experiment to investigate the effect of light intensity on photosynthesis in sun and shade plants.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you **must** use:

- Sun plant
- Shade plant
- Bench lamp with 60 W bulb
- 5 filters (F1, F2, F3, F4, F5) which can be adjusted to allow different amounts of light to pass through
- 1% sodium hydrogencarbonate solution

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware, e.g. a variety of different sized beakers, measuring cylinders, and syringes for measuring volumes
- Forceps
- Timer, e.g. stopwatch

Your plan should:

- have a clear and helpful structure such that the method you use is repeatable by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 14]

This image shows a full page of primary-ruled paper. It features ten sets of horizontal dashed lines, each set consisting of three parallel lines. These lines are evenly spaced vertically across the entire page, providing a guide for handwriting practice. The background is white, and there are no margins or other markings present.

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